

House Price Prediction System Using Ensemble Learning Approach

Abhishek Kumar
Department of CSE
MMHA & PU, Patna
Bihar, India
drabhishekkumarkanth@gmail.com

Paras Jain
School of Computing Science
Engineering and Artificial Intelligence
VIT Bhopal University, Kothrikalan
Bhopal, MP, India
paras.jain@vitbhopal.ac.in

Priyank Pandey
Department of CSE
Graphic Era Deemed to be
University Dehradun, India
priyankpandeyrc@gmail.com

Shyamalendu Das
Humanities Department
SRMCEM, Lucknow, India
shyamalendu.asd@srmcem.ac.in

Vishan Kumar Gupta
Amity School of Engineering and
Technology
Amity University Punjab Mohali, India
vishangupta@gmail.com

Dhiraj Kumar Gupta
Department of Quality Control
Sun Pharmaceutical Industrial Limited
Halol, Vadodara, Gujrat, India
dhirajg2206@gmail.com

Abstract— The real estate industry is among the most active and critically important industries for any economy. An accurate forecast of the prices of houses is very important to the buyers, sellers, investors and the policy makers. In years gone by, other methods of valuation have been used, but new advancements have been made in machine learning meaning the prediction speeds and the level of accuracy have been boosted. In this article, I would like to know how the development of various sophisticated models of ML can be used to forecast house prices: the most used models, the most favorite datasets and informative features, as well as the appropriate evaluation measures. Real estate is a continually expanding business that is prominently involved in the world economy. To precisely estimate house prices is always a daunting task due to various factors that vary from physical to non-physical factors include size, location, market, and economic conditions. Digital advancements especially, the incorporation of machine learning in the analysis of financial markets, housing and other sectors has enabled real estate specialists, banks and other institutions and government bodies analyze trends and develop sophisticated models to predict housing prices more effectively than conventional analytical models. In the context of this article's expanse, we provide a deeper look into the components of house price prediction, with additional information on datasets, feature engineering, specific algorithms, optimization tactics, known difficulties, and future developments.

Keywords— *Housing Price Prediction, Linear Regression, Machine Learning, ANN, Accuracy.*

I. INTRODUCTION

Price forecasts in the past have been based on features such as area, size of the house, and sales in the neighborhood. These methods mostly use simple statistical analysis which may include linear regression. However, due to the existence of a large number of markets and the impact of various factors on the real estate market such as economic conditions, interest rate and neighborhood characteristics, there is need for the development of refined models. More data inputs are acceptable when using machine learning algorithms, and relationships between variables can be non-linear, which produces more accurate forecasting [1].

Traditionally, agents have applied hedonic pricing models which are simple regression models that give estimates of

house prices using features such as location, size and state. Even though these methods are quite simple, there is often no chance to see the nonlinear dependence or the combination of variables. Regarding the capabilities of machine learning methods, they can recognize more sophisticated dependencies in the massive amount of big high-dimensional space. These techniques learn from the data provided; that is, the models are modified when more data is obtained, and the techniques improve with time. In contrast to conventional models, ML models are free form and versatile and thus ideal for application in the fluid real estate market [2].

1.1 Importance of House Price Prediction: Accurate prediction of house prices helps stakeholders in various ways:

- *Homebuyers:* Make better purchasing decisions based on future value expectations.
- *Investors:* Identify potential profit opportunities.
- *Banks and Financial Institutions:* Determine mortgage risks and pricing strategies.
- *Government and Policymakers:* Gain insights into the housing market for regulation and planning purposes.

1.2 Advanced Machine Learning Techniques

A. Linear Regression (Baseline Model): Linear regression continues to be widely used as the initial model in house price prediction because of its plain and basic approach. The model postulates that features (e.g., a number of rooms, area in sq. ft.) are directly proportional to the target variable (price). While it carries the stamp of a classical method for statistical analysis, linear regression forms the basis of more complex models of machine learning [3]. As it relates to the study on house price prediction:

- *Limitation:* Fails to capture non-linear relationships and interactions between features.
- *Strength:* Simple to implement and interpret.
- *Weakness:* Supposes linearity between the independent variables and the target price as opposed to many actual data sets.

B. Decision Trees: Decision trees construct a model that divides data sets into more bifurcated subsets comparing

different features. The decision trees combine a set of if-then statements that are used to divide not only the data but also the features by the values. The last one of them is the forecast of house price. These splits are recursive and are meant to reduce variance in the target variable in order to maximize the accuracy of the prediction [4].

- *Advantages:* They are straightforward to understand and tell others about. The interaction between decision trees and nonlinear and complicated relationship with the existing variables means it happens to be more versatile than linear regression.
- *Limitation:* They are sensitive to overfitting more so when the data set involved is small. ANN models are prone to overfitting of the data in that they learn well from the training data but may not generalize well when trying to learn.

C. Random Forest: Random forest is an example of ensemble learning which generates a multiple number of decision trees and makes an average determination [5]. This makes the decision more accurate and reduces the level of variance that comes with decision trees and is likely to overfit.

- *Overview:* A decision tree reusing model that integrates more than one decision tree. Random forests train many trees from random bootstrap samples of the training data and random subsets of the features and an actual class label is produced by getting a majority of votes from all of the trees.
- *Advantages:* Minimises the problem of building a model that performs well on training phase but poorly on the test phase, compared to single decision trees. Reduced variance or better performance, than when a single decision tree is used.
- *Limitation:* More than one decision tree is hard to interpret as compared to interpreting a single decision tree. Although random forests applied herein effectively address the problem of overfitting, the algorithm may contain sensitive hyperparameters that may need further optimization such as the number of decision trees, the `max_depth` of the trees, and the minimum number of samples required to split an internal node (`samples_per_leaf`).

D. Gradient Boosting Machines (GBM): GBMs like XGBoost and LightGBM have become the go-to models for predictive tasks, including house price predictions. These models build trees sequentially, with each tree correcting the errors of the previous one.

- *Overview:* Gradient boosting is an ensemble method where models are built sequentially, each one improving on the errors of the previous model. Popular algorithms like XGBoost, and LightGBM fall under this category.
- *Advantages:* Handles non-linearity well, effective with large datasets, Captures complex non-linear relationships. It can Outperforms most models in competitions due to its fine-tuning capabilities.
- *Limitation:* Computationally expensive and may overfit if not tuned properly. Gradient boosting models are prone to overfitting, especially if not tuned carefully. They also

require considerable computational resources for large datasets.

E. Artificial Neural Networks (ANNs): Neural networks simulate the human brain by using interconnected nodes (neurons) to process information. For house price prediction, deep learning models can learn complex patterns in the data. Neural networks, particularly deep learning models, are a powerful method for house price prediction when data is vast and complex. ANNs can automatically extract features from raw data, making them less reliant on manual feature engineering [6].

- *Advantages:* Highly adaptive Learns non-linear, complex and high dimensional data patterns and relationships of features, notice conditional dependencies of the features.
- *Challenges:* ANNs take a long time to compute and may have to be trained on large data in order to avoid over training. Further, they require a little less interpretability than the traditional ones such as, linear regression, decision trees, etc. Takes a large amount of data and significant amount of time for calculation of big values.

F. Support Vector Machines (SVM): SVMs are models that are suitable for both regression analysis as well as for use in classification. The ability to map into higher dimensions is an advantage of SVR specially when concerned with non-linear asymptotes when house price is being predicted.

- *Overview:* SVMs are models for classification and regression which seek to determine the best boundary or hyperplane within the data to classify the data sets or predict the result variable.
- *Advantages:* Effective in high-dimensional spaces, Effective in high-dimensional spaces and robust against overfitting, especially when the number of dimensions exceeds the number of observations.
- *Challenges:* Not well-suited for large datasets due to its computational complexity. Not easily scalable to large datasets.

G. K-Nearest Neighbors (KNN): It makes its predictions based on the idea that one will find the 'K' nearest houses that lie closest to the current house in the dataset and then the average price of these 'K' houses serves as the predicted price for the current house. Even though the model is basic and not a parametric method, it serves as the reference model [7].

- *Overview:* KNN is one form of the nearest neighbor methods (no-parametric method) that are used in the model to make predictions by use of K nearest data points used during training.
- *Advantages:* Simple to understand and implement. Easy to implement and understand.
- *Limitation:* Ignores noisy data and not network friendly and does not work well on large stream of data. Not suitable for noisy data set and not so efficient on large data set.

H. Hybrid Models: Some of the hybrid models use several Machine Learning methods at once in order to construct better anticipating models. For instance, Linear regression, Decision tree, GBM, ANN, SVM, KNN, and Random Forests models when combined would be more effective since every model has its recommendation level [8].

- *Advantages:* There are still advantages with these models because each of them can provide different details of the data with higher accuracy and performance.
- *Challenges:* It was elucidated that it is easier to design and implement pure models than the hybrid ones, and the latter requires much effort and many resources for training and fine-tuning.

1.3 Feature Engineering: Beyond Basics:

Feature extraction refers to the idea that raw data inputs must be modified in a way that creates inputs of value to the machine learning algorithm. It can also be appreciated that in house price prediction, many-engineered features considerably improve the model's performance.

A. Geospatial Features

- *Location Data:* Location, location and location (latitude, longitude, and the degree of closeness to facilities such as schools, parks, and other transport facilities) are vital.
- *Neighborhood Data:* The levels of criminality, areas of designation, and quality of the air surrounding a house will determine its price.

B. Temporal Features

- *Time on Market:* The number of days a property has been on the market may indicate demand in the area.
- *Seasonality:* House prices often fluctuate based on the time of year. For example, prices might spike in spring and summer when families are more likely to move.

C. Economic and Financial Data

- *Mortgage Rates:* Interest rates have a direct impact on housing affordability.
- *Inflation and Unemployment Rates:* These macroeconomic indicators can be used to forecast trends in the housing market.

4. *Polynomial Features:* Capturing interaction effects between variables (e.g., interaction between the size of a house and its location) [21].

5. *Dimensionality Reduction:* Techniques like PCA (Principal Component Analysis) can reduce noise by eliminating irrelevant or redundant features [9].

6. *Feature engineering for house price prediction:* Feature engineering is crucial to in-house price prediction [22]. Key features include:

- *Location:* Neighborhood, proximity to schools, public transport, and amenities.
- *Size:* Square footage, number of bedrooms and bathrooms, lot size.
- *Age and Condition:* Year of construction, recent renovations.
- *Economic Indicators:* Interest rates, employment rates, GDP growth.
- *External Factors:* Crime rates, pollution, and natural disaster risk.

II. LITERATURE REVIEW

House price prediction has gained significant attention in recent years due to the increasing importance of data-driven decision-making in real estate. Researchers have explored various machine learning techniques to improve the accuracy of price forecasting, leveraging diverse datasets that include property attributes, location factors, and economic indicators. Studies have demonstrated the effectiveness of algorithms such as linear regression, decision trees, and ensemble methods like random forests and gradient boosting in capturing complex relationships within the data. Additionally, advancements in feature engineering and data preprocessing have enhanced model performance by identifying and selecting the most influential variables. However, challenges remain in addressing biases in data and ensuring model fairness, prompting ongoing discussions about ethical considerations in real estate analytics. Overall, the integration of machine learning in house price prediction presents both opportunities for enhanced accuracy and challenges related to equity and data integrity.

III. DATASETS FOR HOUSE PRICE PREDICTION

Several public datasets can be used for house price prediction:

- Kaggle House Prices Dataset:* A well-known dataset from Ames, Iowa, contains rich data, including 79 explanatory variables covering various aspects of residential homes [10].
- Zillow Housing Data:* Provides comprehensive real estate data with features like sales prices, historical price trends, and geographical information across the U.S.
- UCI Boston Housing Dataset:* A smaller dataset with fewer features but is often used as a benchmark for regression problems in machine learning.
- Private Datasets:* Many real estate companies and financial institutions maintain proprietary datasets, often containing even richer information (e.g., loan records, transaction histories, and buyer preferences).

Ames Housing Dataset is another famous machine learning and data analysis dataset. There is a variety of features and attributes of the residential homes in Ames, Iowa, USA. The dataset is most employed in regression exercises especially for housing price predictions [11]. In this section, the glance of dataset is presented in Table 1 below.

Here are some key details about the Ames Housing Dataset:

- *Number of Instances:* The dataset is record based and has a total of 2,930 instance or observation records.
- *Number of Features:* There are 79 distinct type of features or variables that captures specific characteristics of the residential properties.
- *Target Variable:* The dependent variable in the dataset is named "SalePrice" – it is the price to which houses were sold at.
- *Data Types:* The features include both the number and a categorical number that embraces various aspects like lot size, number of rooms, area, construction, and many more.

IV. MODEL EVALUATION AND METHODOLOGY

A. When evaluating house price prediction models, the following evaluation metrics are commonly used [12]:

- Mean Absolute Error (MAE): An average of the selected measure of the absolute degree of inconsistency between the predicted and actual prices.
- Mean Squared Error (MSE): Attributing higher absolute weights to the larger errors is convenient for outliers' detection [16].

- Root Mean Squared Error (RMSE): The square root of Mean Square Error which gives an interpretable error in the same units as the price [17].
- R-squared (R^2): Offers an indication of how accurately the model predicts the variation of the input variables, house prices [19].
- Accuracy: It is evaluated as the percentage deviation of the actual target to the predicted target with some acceptable error [19].

TABLE I. DATASET DESCRIPTION

PID	MS Sub Class	Lot Frontage	Lot Area	Lot Shape	Land Contour	House Style	Overall Qual	Year Built	Roof Style	Exterior 1st	Exterior 2nd	Total Bsmt SF	Sale Price
526301100	20	141	31770	IR1	Lvl	1Story	6	1960	Hip	BrkFace	Plywood	1080	215000
526350040	20	80	11622	Reg	Lvl	1Story	5	1961	Gable	VinylSd	VinylSd	882	105000
526351010	20	81	14267	IR1	HLS	1Story	6	1958	Hip	Wd Sdng	Wd Sdng	1329	172000
526353030	20	93	11160	Reg	Bnk	1Story	7	1968	Hip	BrkFace	BrkFace	2110	244000
527105010	60	74	13830	IR1	Lvl	2Story	5	1997	Gable	VinylSd	VinylSd	928	189900
527105030	60	78	9978	IR1	Lvl	2Story	6	1998	Gable	VinylSd	VinylSd	926	195500
527127150	120	41	4920	Reg	Lvl	1Story	8	2001	Gable	CemntBd	CmentBd	1338	213500

$$\text{Accuracy} = \frac{100}{n} \sum_{i=1}^n q_i,$$

$$q_i = \begin{cases} 1 & \text{if } \text{abs}(P_i - A_i) \leq \text{err} \\ 0 & \text{otherwise,} \end{cases}$$

where A is the actual target, P is the predicted target, err is some acceptable error, and n is the total number of instances.

B. *Model Optimization Techniques*: Model optimization techniques such as hyperparameter tuning (e.g., Grid Search, Random Search) and cross-validation are essential to improving performance and avoiding overfitting. Cross-validation ensures that the model is tested on multiple subsets of data to generalize well to unseen data. Optimizing machine learning models is essential for achieving the best possible accuracy and generalization [13]. Key techniques include:

1. Hyperparameter Tuning

- Grid Search and Random Search are commonly used to fine-tune hyperparameters like the number of trees in Random Forest or learning rate in GBM models.
- *Bayesian Optimization*: More sophisticated than grid search, Bayesian optimization reduces the time needed to find the optimal hyperparameters by learning from previous evaluations [20].

TABLE II. Hyperparameters Tuning

Model	Required package	Method	Tuning parameters
Random Forest (RF)	randomForest	randomForest	mtry = 2, ntree = 500
Decision Tree (DT)	rpart	rpart	usesurrogate = 0, maxsurrogate = 0
Neural Network (NN)	nnet	nnet	size = 10
Linear Model (LM)	none	lm	method = "qr"

2. *K-Fold Cross-Validation*: Ensures that the model is tested on different subsets of data, minimizing the risk of overfitting and improving generalization [14].

3. *Regularization*: Regularization techniques like Lasso and Ridge regression add penalty terms to the loss function, preventing the model from becoming overly complex [15].

4. *Real-World Challenges*: While machine learning models for house price prediction have improved significantly, several real-world challenges remain:

- *Data Quality*: Real estate data is often noisy or incomplete, requiring extensive cleaning and preprocessing.
- *Economic Volatility*: Sudden economic changes, like a financial crisis, can cause house prices to rise unpredictably.

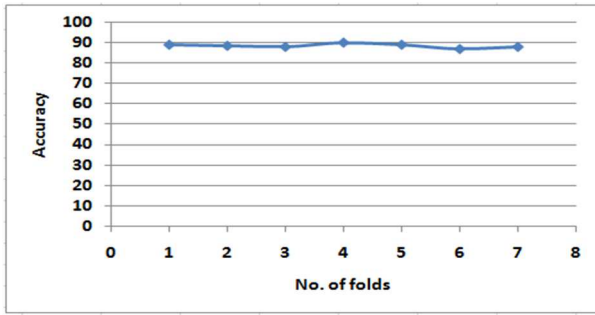


Fig. 1. Robustness of Model under K-folds cross validation

- **Regulatory and Policy Shifts:** Government interventions, like tax changes or housing subsidies, can disrupt trends and make historical data less reliable.
- **Ethical Considerations:** Bias in data (e.g., redlining) can perpetuate inequality if not handled carefully.
- **Integration with Geographic Information Systems (GIS):** Machine learning models will increasingly incorporate real-time geospatial data to improve location-based predictions.

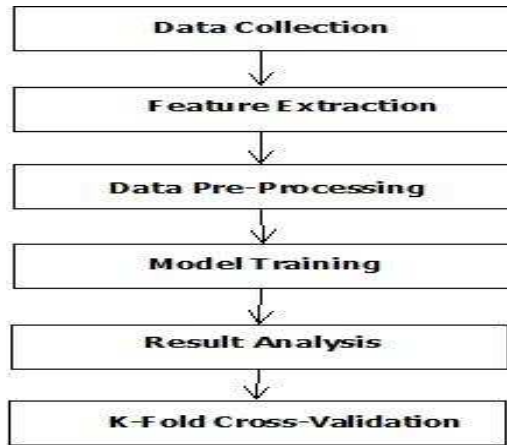


Fig. 2. Methodology

V. RESULT ANALYSIS AND COMPARISON

The distribution of data in the training–testing experiment has been set to 70% and 30%, respectively. RMSE, correlation (r), R^2 , and accuracy are the performance metrics of our proposed multilevel ensemble model during the

regression phase. Table 2 shows the comparative performance of various models. The results show that our proposed ensemble model has outperformed the other existing machine learning models.

Figure 3 is showing the performance of proposed hybrid ensemble model, where we can see that our proposed ensemble model is performing well in comparison to discussed existing model. Table 3 is also same in term of performance metric correlation, R-squared, RMSE, and Accuracy.

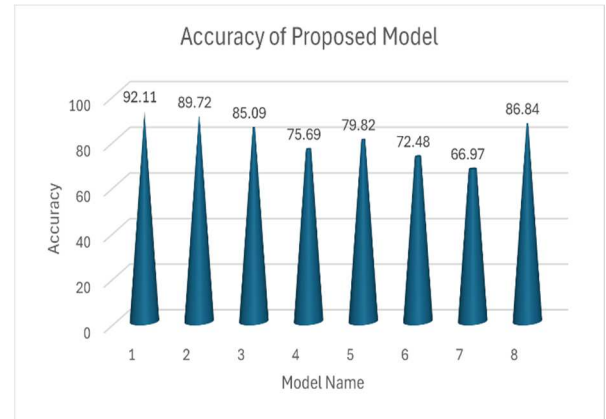


Fig.3. Comparison of Our proposed Model with Existing One

TABLE III. Performance metric of proposed model

Techniques	Coorealtion	R-Square	RMSE	Accuracy
Proposed ensemble model	0.92	0.85	3.48	92.11
Random Forest	0.94	0.88	4.57	89.72
Decision Tree	0.90	0.81	3.69	85.09
SVM	0.62	0.38	8.03	75.69
Linear Regression	0.84	0.71	5.54	79.82
KNN	0.49	0.24	10.45	72.48
GBM	0.68	0.46	67.32	66.97
ANN	0.91	0.83	3.95	86.84

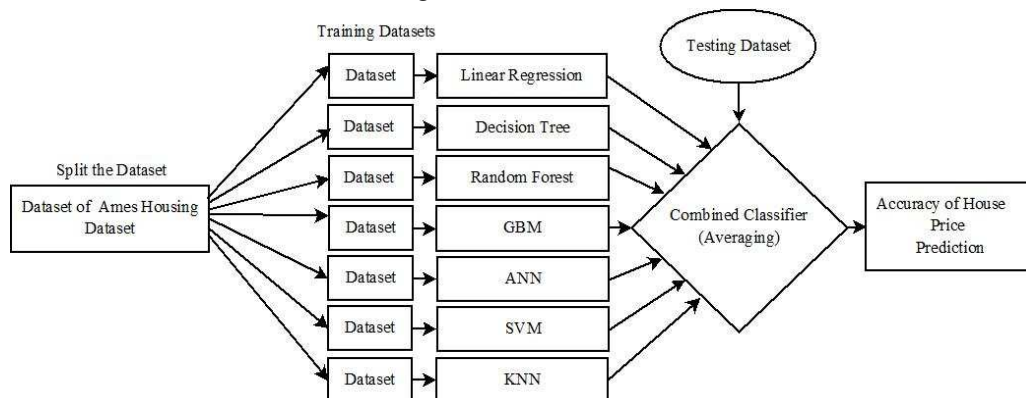


Fig. 4. Hybridization of Model

VI. CONCLUSION

Advanced machine learning approaches like gradient boosting, random forests, and neural networks offer a powerful toolkit for predicting house prices with greater accuracy than traditional models. With the right features, datasets, and model optimization techniques, machine learning can significantly enhance predictions in the real estate market, benefiting a wide range of stakeholders. As data availability and computational power continue to grow, the potential for machine learning in real estate will only expand further. Our proposed ensemble model is evaluated on different performance parameters, and compared with other peer models for regression phase, individually. Through the intensive experiments and comparisons, it concludes that our proposed model, in spite of having highly imbalanced data, outperformed other examined methods in both its levels and its performance is almost consistent in k-fold cross-validation. Finally, to prove the validity of the proposed ensemble model, we have tested it on some new houses, which are neither part of training and testing dataset, where it performed very well.

VII. FUTURE SCOPE

When we talk about its future scope then integration of AI and Big Data for house price prediction, Impact of sustainability and user driven approach in this area, and the role of advanced analytics and real-time data processing are missing. Addressing these areas not only improves the accuracy and usability of house price prediction systems but also aligns them with contemporary trends in sustainability and user engagement. Integrating advanced analytics, considering environmental impacts, and enabling user-driven customization can significantly enhance the predictive capabilities and market relevance of these systems.

REFERENCES

- [1] "House Price Index. Federal Housing Finance Agency," <https://www.fhfa.gov/>, accessed on September 1, 2019.
- [2] C. Fan, Z. Cui, X. Zhong, "House Prices Prediction with Machine Learning Algorithms," Proceedings of the 2018 10th International Conference on Machine Learning and Computing - ICMLC 2018. doi:10.1145/3195106.3195133.
- [3] T.D. Phan, "Housing Price Prediction Using Machine Learning Algorithms: The Case of Melbourne City, Australia," 2018 International Conference on Machine Learning and Data Engineering (ICMLDE), 2018, doi:10.1109/icmlde.2018.00017.
- [4] J. Mu, F. Wu, A. Zhang, "Housing Value Forecasting Based on Machine Learning Methods," Abstract and Applied Analysis 2014, 2014:pp. 1–7. doi:10.1155/2014/648047.
- [5] S. Lu, Z. Li, Z. Qin, X. Yang, RSM Goh, "A hybrid regression technique for house prices prediction," 2017 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM) 2017. doi:10.1109/ieem.2017.8289904.
- [6] I. V. Ivanov, "GitHub 2016," <https://github.com/vecxoz/vecstack>, accessed on June, 2019.
- [7] D. H. Wolpert, "Stacked generalization," Neural Networks 1992, vol. 5, pp. 241–59. doi:10.1016/s0893-6080(05)80023-1.
- [8] Q. Qiu "Housing price in Beijing," Kaggle 2018. <https://www.kaggle.com/ruiqurp/lianjia/>, accessed June 1, 2019.
- [9] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, "Scikit-learn: Machine Learning in Python," The Journal of Machine Learning Research, vol. 12, 2011, pp: 2825–2830.
- [10] L. Breiman, "Random Forests," Springer Link. <https://doi.org/10.1023/A:1010933404324>, accessed on September 11, 2019.
- [11] S. Raschka, V. Mirjalili, "Python machine learning: Machine learning and deep learning with Python," scikit-learn, and TensorFlow. 2nd ed. Birmingham: Packt Publishing, 2017.
- [12] T. Chen, C. Guestrin. "XGBoost. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining – KDD, vol. 16, 2016.
- [13] DMLC, xgboost. GitHub, <https://github.com/dmlc/xgboost>, accessed on June 2019.
- [14] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, "LightGBM: A Highly Efficient Gradient Boosting Decision Tree, Advances in Neural Information Processing Systems, vol. 30, 2017, pp:3146–3154.
- [15] Microsoft. LightGBM. GitHub, <https://github.com/microsoft/LightGBM>, accessed June 1, 2019.
- [16] M. K. Singh, P. Singh, V. K. Gupta, K. Mishra, and A. Gupta, "Performance Analysis of CNN Models with Data Augmentation in Rice Diseases," 2023 3rd Asian Conference on Innovation in Technology (ASIANCON), Ravet IN, India, 2023, pp. 1-5.
- [17] S. K. Mishra, A. Gupta, A. Tomar and V. K. Gupta, "Health Care Prediction for Various Diseases using Computational Intelligence Approaches: A Review," 2023 World Conference on Communication & Computing (WCONF), RAIPUR, India, 2023, pp. 1-6.
- [18] K. Fujioka, "A model for simulating residential location choices of working women considering compact city planning," International Journal of Grid and Utility Computing, vol. 15, no.5, 2024, pp. 442-454.
- [19] V. K. Gupta, P. Jain, D. Anand, G. Dhiman, S. Vishwakarma and A. Aeron, "A Credit Risk Assessment System for Financial Institutions Utilizing Deep Learning," 2024 1st International Conference on Advanced Computing and Emerging Technologies (ACET), Ghaziabad, India, 2024, pp. 1-6.
- [20] V. Rawat, P. Jain, A. Tomar, P. Guzar, V. Negi and V. K. Gupta, "A Web-Based Real Time Chat Application for User Friendly Interaction," 2024 1st International Conference on Advanced Computing and Emerging Technologies (ACET), Ghaziabad, India, 2024.
- [21] V. Panwar, P. Jain, A. Tomar, V. K. Gupta, H. Tiwari and V. Shukla, "Automatic Face Recognition-Based Attendance System using 2D Convolutional Neural Network," 2024 1st International Conference on Advanced Computing and Emerging Technologies (ACET), Ghaziabad, India, 2024, pp. 1-6.
- [22] A. Tomar, P. Jain, V. Panwar, V. K. Gupta, V. Shukla and Vidisha, "A Unified Communication Model for Multiple Sensory Disabilities Using Convolutional Neural Networks," 2024 1st International Conference on Advanced Computing and Emerging Technologies (ACET), Ghaziabad, India, 2024, pp. 1-5.